Practice Paper 1(Module 1)

1. How massage passing helps in inter process communication, explain with diagram.
2. Draw a diagram for process scheduling, what is the use of queue in process scheduling.
3. What is Multiprocessing and multitasking?
4. How a conventional operating system is differing from real-time operating system?
5. How aging will help in priority scheduling technique?
6. Consider the set of 5 processes whose arrival time and burst time are given below-

Process Id	Arrival time	Burst time
P1	3	1
P2	1	4
P3	4	2
P4	0	6
P5	2	3

If the CPU scheduling policy are SJF non-preemptive/Preemptive and Round robin (TQ=2), calculate the average waiting time and average turnaround time.

7. What is OS Kernel. Explain with diagram.
8. How layered architecture will helpful to maintain the security by operating System?

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Mid-Term Examination
Even Semester, 2018-19

Programme: B. Tech **Branch: CSE** **Year: II**
Subject & Code: Introduction to Microprocessor (BCSC0007)
Time: 1 Hour Maximum Marks: 15

Section – A

3 x 2 = 6 Marks

Q1. Define PSW? Distinguish between 8085 and 8086 microprocessors?

Q2. Draw the 8086-flag register bits pattern. Explain flags which are not available in 8085 microprocessors.

Q3. Define stack & subroutine in microprocessor. Calculate the maximum delay by using microprocessor register as a counter with clock frequency 2MHz.

Section – B

3 x 3 = 9 Marks

Q1. Explain CMP instruction with example. Write Assembly language program to find max number from a list starting from memory location 3000h and number of items in given list store at memory location 2500h.

Q2. Draw the functional block diagram of 8086. List all functional blocks of execution unit and describe bus interface unit in brief.

Q3. How many times the loop will be executed (write in decimal)? Find out total t-state for each loop.

a.
```
    MVI B, 11h
L1: DCR B
    JNZ L1
```

b.
```
    LXI H, 1000h
L1: DCR H
    MOV A, L
    ORA H
    JNZ L1
```

c.
```
    MVI C, 00h
L1: DCR C
    JNC L1
```

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Mid-Term Examination, Even Semester 2020-21

B.Tech, Year: 2nd, Semester: 4th

Introduction to Microprocessors(BCSC 0007)

Time : 2 Hours **Maximum Marks: 30**

Section- A

Note: Attempt All Three Questions. 3 x 2 = 6 Mark

I. What is the use of RESET IN and RESET OUT instruction.
II. Write any four Logic and Bit Manipulation Instructions.
III. Explain Register Indirect Addressing Mode with example.

Section- B

Note: Attempt All Three Questions. 3 x 3 = 9 Marks

I. How AD7-AD0 signal lines are used to transfer both data and low order address using multiplexing? Explain with diagram.
II. Write a program to exchange the contents of memory locations 3000H and 4000H.
III. Explain the following instructions with example:
   1. LXI
   2. SUI
   3. INR

Section – C

Note: Attempt Any Three Questions. 3 x 5 = 15 Marks

I. Draw Timing diagram for the execution of the instruction MVI B,44.(HEX code of instruction MVI B is 06 which is stored at memory location 2001and data 44 is stored at memory location 2002).

II. Explain all the Control and Status Signals of microprocessor 8085.

III. Write a program in microprocessor 8085 assembly language for addition of two 16-bit numbers. Add the 16-bit number in memory locations 4000H and 4001H to the 16-bit number in memory locations 4002H and 4003H. The most significant eight bits of the two numbers to be added are in memory locations 4001H and 4003H. Store the result in memory locations 4004H and 4005H with the most significant byte in memory location 4005H.

IV. What will be the value of all 5 Flag registers and Accumulator after executing each program:

1. MVI A, 86
   MVI B, A7
   ADD B

2. MVI A, 59
   MVI B, 99
   ADD B

3. MVI A, EF
   MVI B, CD
   ADD B

4. MVI A, 79
   MVI B, 5A
   SUB B

5. MVI A, 86
   MVI B, 97
   SUB B

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First Mid-Term Examination, Even Semester 2019-20

B. Tech (CSE, IOT, DA, CCV & CSF), II, IV

BCSC 0007: Introduction to Microprocessor

Time: 1 Hour

Maximum Marks: 15

Section – A

Attempt All the questions. 3 x 2 = 6 Marks

1. Write any four features of 8085 microprocessor? List four applications of Microprocessor?
2. Draw and explain flag register of 8085 MPU. What will be the value of all flags (except auxiliary carry flag) of 8085 after performing subtract 8E by 7E hexadecimal data.
3. Write the working of following pins of 8085 MPU.

   I. HOLD and HLDA II. $\overline{\text{RESET IN}}$ and RESET OUT

Section – B

Attempt All the questions. 3 x 3 = 9 Marks

1. Define Instruction cycle, machine cycle. The instruction MVI B, FF (hexadecimal) is useful to load FF (hexadecimal) as data byte in register B. It is a 2-Byte instruction. Draw the timing diagram for MVI B, FF (hexadecimal).
2. Write the assembly language program to subtract two sixteen bits numbers available in BC and DE register pair, after performing operation store the result in 3000 and 3001 memory location respectively. Also write description of each instruction with program.
3. List all addressing modes used in 8085 MPU. Explain any two with example.

Microprocessor Assignment Questions

1. A microprocessor uses eight address lines for accessing a memory. What is the maximum size of the memory that can be directly addresses by this microprocessor?

2. What do you understand by the statement Intel 8085 is an 8-bit microprocessor but its address bus is of 16-bit.

3. Explain the internal architecture of Intel 8085 microprocessor with the help of neat and clean block diagram.

4. Discuss the function of SP, PC and SF in 8085 microprocessors. What will be the content of status flag after the operation ADD CC and E8.

5. Draw the pin diagram of 8085 microprocessor and explain the functions of various pins.

6. Classify Intel 8085 instructions in various groups giving suitable examples.

7. Explain the following Intel 8085 instructions with examples:
   a. MVI
   b. STA
   c. INX
   d. IN
   e. ADC
   f. ANI
   g. RRC

8. Write instructions to store the contents of register B (42H) in the memory location 2150H using the instruction MOV, STAX and STA.

9. Explain what operation will take place when the following instructions are executed by taking one example:
   a. CMA
   b. NOP
   c. ANA
   d. RAL
   e. POP
   f. SIM
   g. RIM

10. Discuss the different addressing modes of Intel 8085 microprocessor.

11. What is the purpose of providing registers in a microprocessor? Describe the various registers which are usually provided in a microprocessor.

12. What do you mean by instruction word size? Give example.

13. Write an assembly language program (ALP) for the 2'S complement of an 8-bit number.

14. Write an assembly language program (ALP) to place the content of memory location in a register and that of register to any memory location. Choose the memory location and the register.

15. In the following, after POP PSW is executed, what are the contents of the 8085 Processor Status Word (PSW) and Accumulator (A)? Assume that [20C2] = 01and [20C3] = 05.

```
LXI  SP, 20C2 H
POP PSW
```

16. In the following instruction sequence, find the contents of the HL register pair after the execution of the instruction DAD B. Assume [2000] = 01 H and [2001] = 00 H.

```
LXI   B, 2000 H
LHLD  2000 H
DAD   B
HLT
```

17. Discuss the instruction cycle, machine cycle and state.

18. Draw the timing diagram for the instruction LDA C201H

19. Mention the number and name of the different machine cycles in the following instructions:

    a. LXI H, 2000 H
    b. MOV A, M
    c. STA 2000 H

20. Differentiate between the following instructions:
    a. RAL and RLC
    b. CMP and SUB
    c. SHLD and LHLD
    d. NOP and HLT

21. What is the function of DAA instruction? Write an assembly language program (ALP) for 8085 that adds decimal numbers 345 and 753.

22. What will be the contents of the status flags after the execution of different compare instructions in the following sequence of Intel 8085 instructions. Assume [A]=6D H and [0120] = 49 H.

Memory Address	Hex Codes	Mnemonics	Operand	Comment
2000 H	21	LXI	H, 0120H	[H-L]← 0120
2001 H	20			
2002 H	01			
2003 H	BE	CMP	M	[A] – [[H] [L]]
2004 H	FE	CPI	6F H	[A] – 6F
2005 H	6F			
2006 H	FE	CPI	6D H	[A] – 6D
2007 H	6D			
2008 H	C3	JMP	2008 H	
2009 H	08			
200A H	20			

23. Write a set of instructions to alter the contents of flag register in 8085.

24. (a) Pack the two unpacked BCD numbers stored in memory locations 3200H and 3201H and store result in memory location 3300H. Assume the least significant digit is stored at 3200H.

(b) Two-digit BCD number is stored in memory location 4200H. Unpack the BCD number and store the two digits in memory locations 3300H and 3301H such that memory location 3300H will have lower BCD digit.

25. Calculate the sum of series of numbers. The length of the series is in memory location 3200H and the series begins from memory location 3201H. Consider the sum to be 8 bit number. So, ignore carries. Store the sum at memory location 3300H.